

Spatial Navigation using Grid Cells: Path Integration in Continuous Attractor Networks

NX-465 Computational Neuroscience: Neuronal Dynamics Mini-Project MP2 — Spring 2026

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Description

Grid cells, first discovered in the medial entorhinal cortex (MEC) of rodents by Hafting et al. (2005) [1], fire in a striking hexagonally periodic pattern as an animal navigates through its environment. Crucially, this spatial code persists even in complete darkness, demonstrating that grid cells do not merely report incoming sensory input, but actively integrate the animal’s self-motion cues (velocity and head direction) into a continuously updated internal position estimate—a fundamental cognitive mechanism known as **path integration**.

In this mini-project, we build a **Continuous Attractor Network (CAN)** model of grid cells, closely following the seminal framework established by Burak and Fiete (2009) [2]. Attractor networks are dynamical systems that constrain high-dimensional population activity into a **low-dimensional neural manifold** (e.g., a 1D ring or a 2D torus) of minimal-energy states, enabling the maintenance of stable representations over time. All simulations use **Forward Euler** numerical integration with $\Delta t = 1$ ms and a rectified linear unit (ReLU) transfer function $f(x) = \max(0, x)$.

Exercise 0 — Getting Started: 1D Ring Attractor

Model equations

The activity profile $s(x, t)$ of a neuron at position x in a 1D ring ($M = 100$) evolves as:

$$\tau \frac{\partial s(x, t)}{\partial t} = -s(x, t) + I^{rec}(x, t) + B(x, t), \quad (1)$$

with $\tau = 10$ ms, constant drive $B = 1.0$, and firing rate $r(x, t) = \max(0, s)$. The recurrent input I^{rec} is a spatial convolution with a Mexican-hat (difference of Gaussians) kernel:

$$I^{rec}(x, t) = \int W(x - x') r(x', t) dx', \quad W(x) = A_{exc} e^{-x^2/\sigma_{exc}^2} - A_{inh} e^{-x^2/\sigma_{inh}^2}. \quad (2)$$

Ex 0.2 & 0.4 — Kernel and ring attractor dynamics

Using $A_{exc} = 0.3$, $\sigma_{exc} = 5.0$, $A_{inh} = 0.2$, $\sigma_{inh} = 10.0$; integrated over $T = 500$ ms from uniform noise $s(x, 0) \sim \text{Unif}(0, 0.1)$:

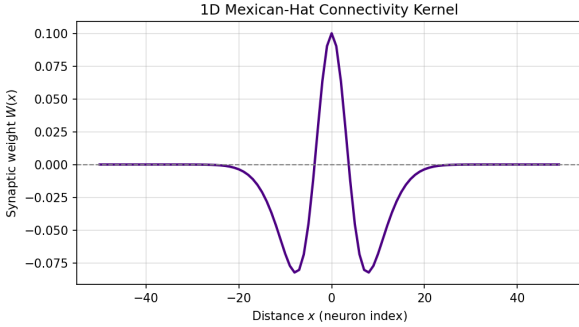


Figure 1: 1D Mexican-hat kernel $W(x)$. Short-range excitation (positive lobe) and long-range inhibition (negative flanks) drive spontaneous pattern formation.

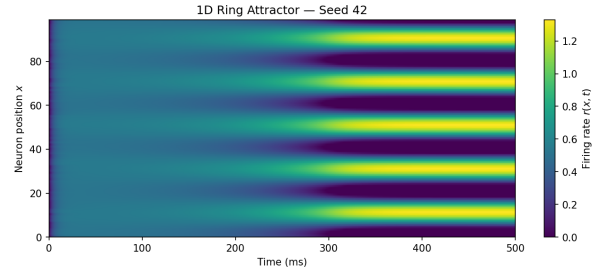


Figure 2: Spatiotemporal firing rate $r(x, t)$ (seed 42). A stable activity bump emerges from uniform noise within ~ 50 ms, demonstrating bump attractor formation.

Ex 0.3 — Biological interpretation

The Mexican-hat profile reflects local recurrent excitation between nearby pyramidal cells (width σ_{exc}) flanked by broader lateral inhibition from interneurons projecting over larger distances ($\sigma_{inh} > \sigma_{exc}$). This center-surround structure suppresses uniform background activity while amplifying spatially periodic patterns, driving the spontaneous formation of a localised activity bump.

Ex 0.5 — Dependence on initialisation

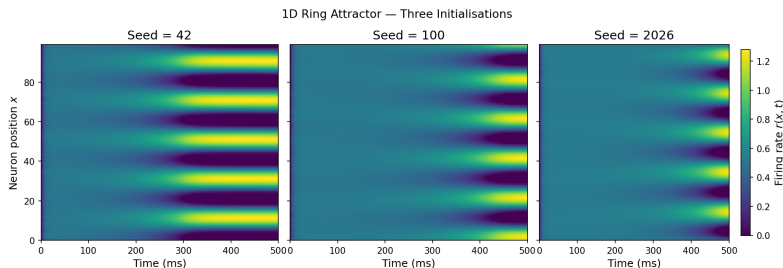


Figure 3: Ring attractor dynamics under three random seeds. Bumps share identical amplitude and width (set by the kernel) but differ in final spatial location.

Observation. The macroscopic bump is robust to initial noise, but its spatial phase is entirely set by initial conditions via spontaneous symmetry breaking. This defines a **continuous attractor**: a manifold of translationally equivalent marginally stable fixed points that supports persistent spatial memory.

Exercise 1 — 1D Velocity Integration

Architecture

For 1D path integration, $M = 100$ neurons split into Right-shifting (R) and Left-shifting (L) sub-populations with kernels offset by $l = 1.0$:

$$W_R(x - x') = W(x - x' - l), \quad W_L(x - x') = W(x - x' + l). \quad (3)$$

Velocity $v(t)$ asymmetrically modulates the feedforward drive $B_0 = 1.5$, $\alpha = 0.5$:

$$B_R = B_0(1 + \alpha v(t)), \quad B_L = B_0(1 - \alpha v(t)), \quad I^{rec} = \int W_{RR} dx' + \int W_{LL} dx'. \quad (4)$$

Ex 1.1 — Bump drift under constant velocity

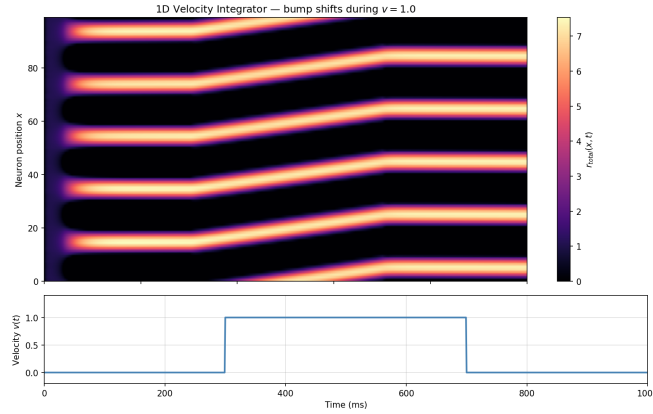


Figure 4: **Top:** Total firing rate $r_{total} = r_R + r_L$. During $v = 1.0$ (300–700 ms) the bump drifts right; it halts at its new position when $v = 0$ is restored. **Bottom:** Velocity profile $v(t)$.

Ex 1.2 — Mechanism of velocity-driven shifting

When $v > 0$, the R population receives a stronger drive ($B_R > B_L$). The offset kernel W_R projects maximal excitation ahead of the bump by l , biasing recurrent input rightward while inhibiting the left flank via the Mexican-hat surround. The drift speed is set by $B_R - B_L = 2\alpha B_0 v$; negative v symmetrically drives leftward translation.

Ex 1.3 & 1.4 — Varying velocities and linear speed–velocity relationship

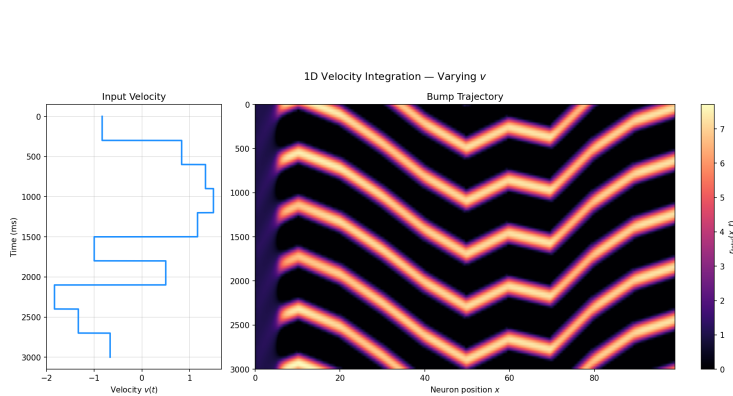


Figure 5: **Left:** Time-varying $v(t)$ (10 blocks of 300 ms from $[-2, -0.5] \cup [0.5, 2]$). **Right:** Bump trajectory $r_{total}(x, t)$, tracking the input with speed $\propto |v|$ and wrapping around the ring.

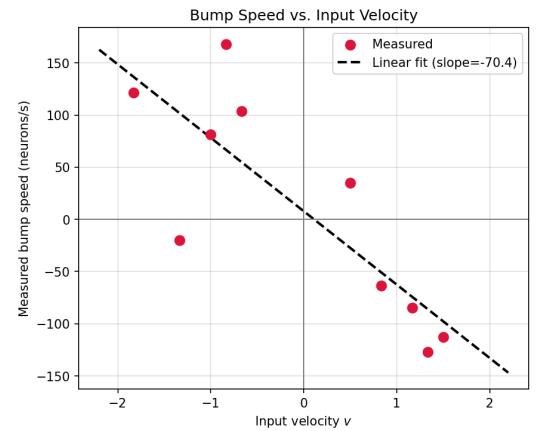


Figure 6: Measured bump speed vs. injected velocity. The **linear** relationship (dashed fit) passing close to the origin confirms continuous velocity integration.

The bump moves at a speed proportional to $v(t)$, instantly reversing direction upon each velocity switch, while its shape remains stable. It wraps around the ring boundaries without disruption, confirming the periodic topology.

The bump’s cumulative displacement tracks the time-integral of velocity, demonstrating that the network performs path integration. The bump position is tracked via the phase of the first Fourier mode of the activity pattern. After phase unwrapping, the bump speed is obtained by finite differencing. Plotting measured speed against input velocity v reveals a clear linear relationship (slope ≈ -70.4 neurons/s), confirming that the network faithfully integrates the velocity input. The fitted line does not pass exactly through the origin, which likely reflects finite-time averaging, small transient effects, and numerical/discretization errors in the phase-based velocity estimate rather than a true failure of the integrator. The negative sign of the slope is a measurement convention: the Fourier estimator $e^{-i2\pi x/M}$ assigns decreasing phase to rightward bump displacement, so the measured speed is the negative of the physical drift speed but the magnitude and linearity are unaffected by this.

Exercise 2 — Extending to 2D Grid Cells

2D architecture and kernel

The Mexican-hat kernel generalizes to 2D via $\|\mathbf{x}\|$:

$$W(\mathbf{x}) = A_{exc} e^{-\|\mathbf{x}\|^2/\sigma_{exc}^2} - A_{inh} e^{-\|\mathbf{x}\|^2/\sigma_{inh}^2}. \quad (5)$$

Four directional populations (N, S, E, W) use kernels shifted by $\pm l$ along each axis. Their feedforward drives are modulated by velocity $\mathbf{v} = (v_x, v_y)$:

$$B_N = B_0(1 + \alpha v_y), \quad B_S = B_0(1 - \alpha v_y), \quad B_E = B_0(1 + \alpha v_x), \quad B_W = B_0(1 - \alpha v_x), \quad (6)$$

with $I^{rec}(\mathbf{x}, t) = \sum_{p \in \{N, S, E, W\}} \iint W_p(\mathbf{x} - \mathbf{x}') r_p d\mathbf{x}'$. Parameters: $\tau = 10$ ms, $l = 1.0$, $\alpha = 0.1$, $B_0 = 1.0$, $M = 40$.

Ex 2.1 & 2.2 — 2D kernel and spontaneous hexagonal grid

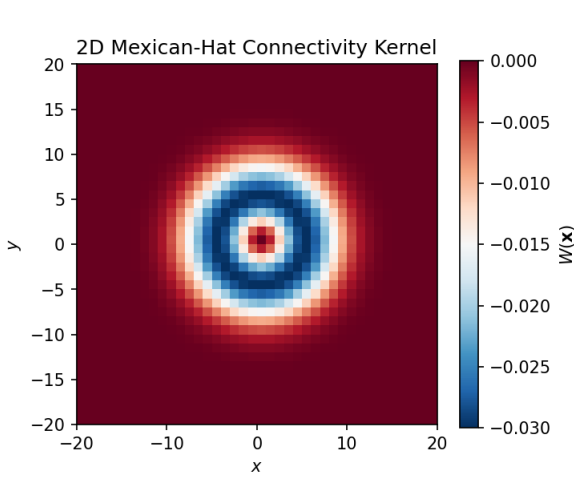


Figure 7: 2D Mexican-hat kernel ($A_{exc} = A_{inh} = 1.0$, $\sigma_{exc} = 4.8$, $\sigma_{inh} = 5.0$). The narrow σ margin sets the spatial frequency of the emergent lattice.

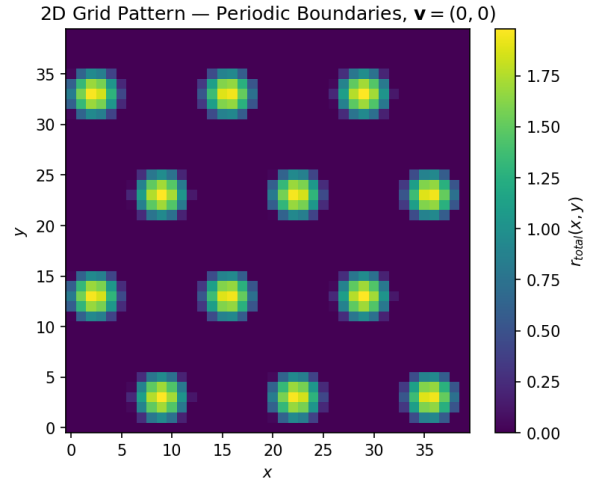


Figure 8: Total firing rate $r_{total}(x, y)$ after $T = 400$ ms ($\mathbf{v} = 0$, torus BC). A stable **hexagonal lattice** emerges spontaneously, directly analogous to biological grid cell firing fields.

Yes, a hexagonal grid pattern is clearly visible. The 2D Mexican-hat kernel destabilizes the uniform background and selectively amplifies the preferred spatial frequency set by $\sigma_{exc}/\sigma_{inh}$, driving the network to its minimal-energy hexagonal configuration.

Ex 2.3 — Velocity integration in 2D

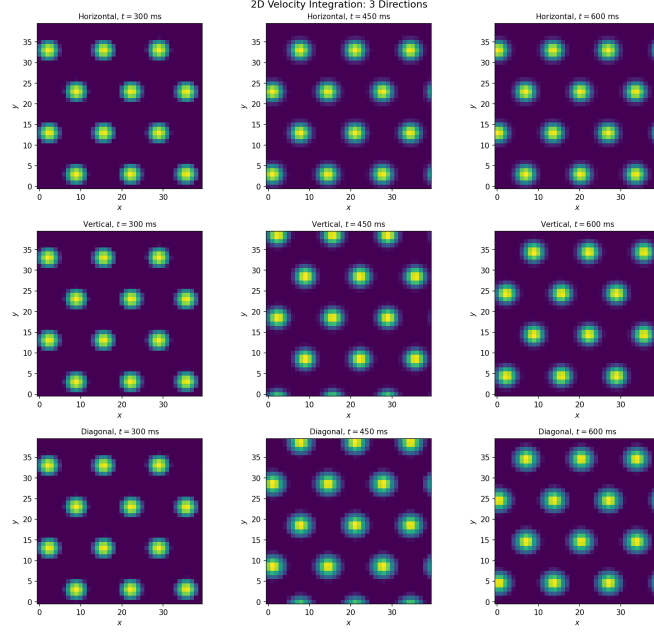


Figure 9: $r_{total}(x, y)$ at $t = 300, 450, 600$ ms for horizontal ($v_x = 2$), vertical ($v_y = 2$), and diagonal ($v_x = v_y = 2$) motion. The hexagonal pattern is *structurally preserved*; only its spatial phase shifts, encoding the time-integrated position $\phi(t) = \phi(0) + \kappa \int_0^t \mathbf{v} dt'$.

The hexagonal pattern translates rigidly in the direction of the applied velocity, with speed proportional to $|\mathbf{v}|$, while its spacing and orientation remain unchanged. This confirms that the network performs 2D path integration: the velocity bias asymmetrically modulates the feedforward drives across the four populations, and the shifted kernels propagate this bias into a rigid lattice translation.

Ex 2.4 (Bonus) — 2D velocity tracking under time-varying input

A time-varying velocity signal is generated by sampling a new random direction and speed in $[1, 2]$ every 100 ms. These velocity components are then used as input to the 2D periodic network.

To estimate the motion of the grid pattern, we use a simple bump-tracking method after an initial stabilization period of 300 ms. We first identify one bright bump in the activity map, then follow its position over time by looking in a small neighbourhood around its previous location. The local centre of mass of this neighbourhood provides a smooth estimate of the bump position, and the velocity is obtained by differentiating this position over time.

This gives a simple qualitative measure of how the moving grid pattern responds to the changing input velocity, while keeping the analysis easy to interpret.

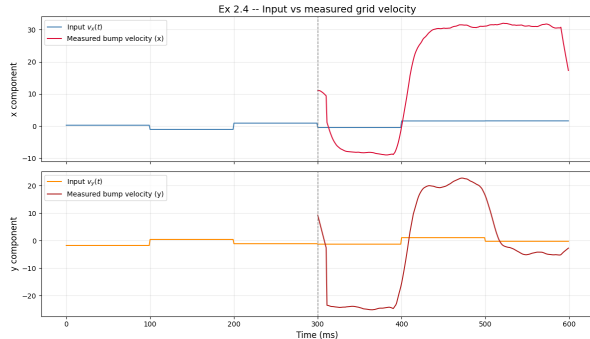


Figure 10: Time series of the input velocity components and the measured motion of the tracked bump in the periodic network. The velocity estimate is obtained after a stabilization period ($t = 300$ ms) and follows the main changes in the input over time, with visible transient delays after each change.

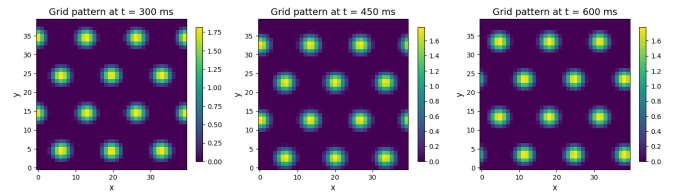


Figure 11: Snapshots of the 2D grid pattern at $t = 300, 450, 600$ ms in the periodic network with time-varying velocity input. The hexagonal lattice remains clearly visible while its position shifts over time, indicating that the pattern mainly translates while preserving its overall structure.

Observation. After the initial stabilization period, the grid pattern moves when the input velocity changes, and

the measured motion follows the main trends of the input in both components. Short transient delays are visible after each change in velocity. The snapshots show that the hexagonal lattice structure is preserved overall, with the pattern mainly translating while keeping a similar spacing and orientation.

Exercise 3 — Aperiodic Boundaries

Ex 3.1 & 3.2 — Grid breakdown and radial masking envelope

Biologically realistic boundaries truncate the convolution (via `mode='constant'`), removing the inhibitory surround for peripheral neurons:

$$I^{rec}(\mathbf{x}, t) = \int_0^M \int_0^M W(\mathbf{x} - \mathbf{x}') r d\mathbf{x}', \quad (7)$$

causing unbounded boundary excitation. A radial envelope $e(\mathbf{x})$ corrects this:

$$e(\mathbf{x}) = \exp(-\gamma^2 \max(0, \|\mathbf{x}\| - R_{fade})^2), \quad (8)$$

with $\gamma = 0.08$, $R_{fade} = f_{ratio} \times M/2 = 5$ ($f_{ratio} = 0.2$, $M = 50$).

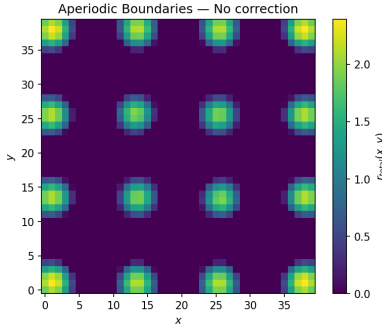


Figure 12: Without correction: boundary artefacts destroy the hexagonal lattice. Peripheral neurons accumulate uncompensated excitation.

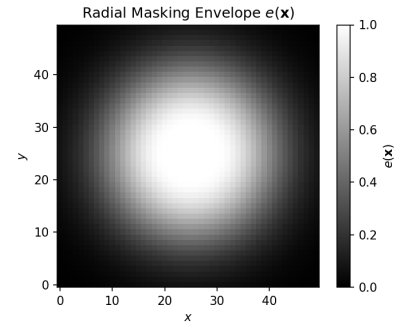
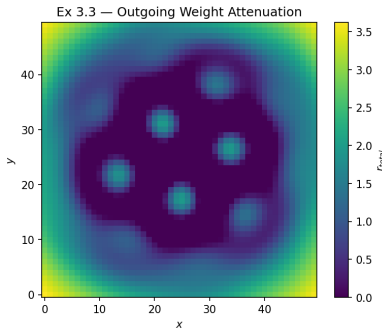


Figure 13: Radial masking envelope $e(\mathbf{x})$ on a 50×50 lattice ($e = 1$ at center, decays to 0 beyond $R_{fade} = 5$).

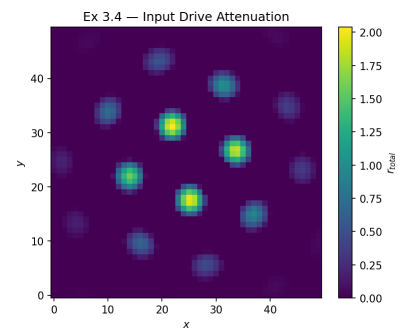
Observation. The truncation of the continuous attractor sheet structurally removes the long-range inhibitory surround for boundary neurons. This asymmetric loss fundamentally disrupts the center-surround competition mandated by the Mexican-hat kernel. Consequently, peripheral neurons receive a net recurrent excitatory current that permanently drives their activity above threshold, saturating the network edges and suppressing the interior grid pattern.

Ex 3.3 & 3.4 — Weight and input attenuation

Weight attenuation (Ex 3.3) scales firing rates by $e(\mathbf{x})$ *before* the recurrent convolution, reducing boundary neurons' efferent projections. **Input attenuation** (Ex 3.4) instead multiplies the feedforward drives $B_{N,S,E,W}$ by $e(\mathbf{x})$, suppressing boundary activity while leaving recurrent weights intact.



(a) Weight attenuation (Ex 3.3).



(b) Input attenuation (Ex 3.4).

Figure 14: Both strategies restore the spontaneous hexagonal lattice in the central unattenuated region ($\approx 40 \times 40$), while boundary neurons remain sub-threshold.

Ex 3.5 — Comparison and biological implications

Table 1: Comparison of boundary correction strategies.

	Ex 3.3 — Weight attenuation	Ex 3.4 — Input attenuation
What is scaled	Efferent rates $r(\mathbf{x}, t)$ before convolution.	Feedforward drives $B_{N,S,E,W}$ at each neuron.
Mechanism	Reduces recurrent projections of peripheral neurons.	Reduces external drive without altering recurrent connectivity.
Pattern quality	Lattice restored centrally; boundary neurons silent.	Lattice restored centrally; boundary neurons have partial activity.
Biological analogue	Reduced efferent arborization at MEC borders.	Reduced thalamic/hippocampal afferents or lower stellate cell excitability.

Both methods restore the static hexagonal lattice, but their dynamical implications diverge. Weight attenuation breaks the continuous translation invariance of the recurrent manifold, inducing grid distortion during locomotion. Input attenuation preserves the symmetric recurrent weights and, as formally shown by Burak and Fiete [2], is necessary and sufficient for undistorted path integration during velocity integration (Ex 3.6).

Ex 3.6 (Bonus) — Velocity tracking under aperiodic boundaries

Using the input-attenuation model from Ex 3.4 with $M = 50$, we apply the same type of time-varying velocity signal as in Ex 2.4. To obtain a simple measure of the pattern motion, we track the position of the activity pattern after an initial stabilization period and estimate its velocity directly from the displacement over time.

This provides a qualitative way to compare velocity tracking in the aperiodic case with the periodic case, while keeping the analysis simple and consistent with the rest of the notebook.

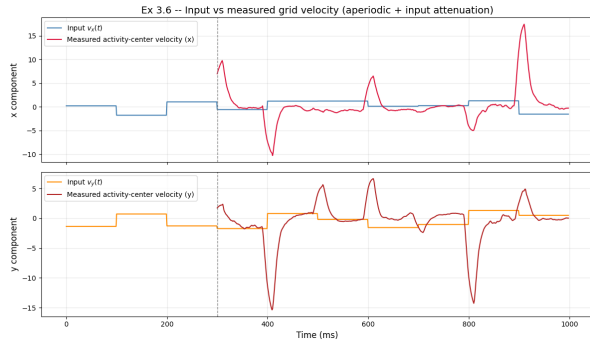


Figure 15: Input and measured grid velocity for the aperiodic network with input attenuation. The velocity estimate is obtained by tracking the motion of the activity pattern after 300 ms, once the grid has stabilized. The measured response captures the main changes in the input, but remains noisier and less accurate than in the periodic case.

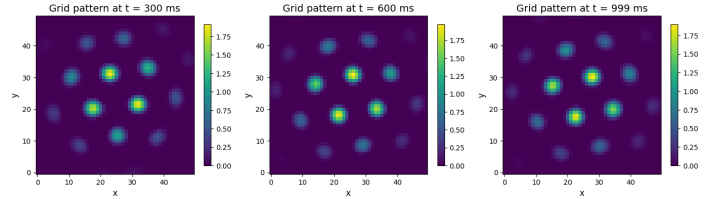


Figure 16: Grid snapshots at $t = 300, 600$, and 999 ms for the aperiodic network with input attenuation. A grid-like pattern remains visible in the centre, while activity near the boundaries stays weak because the input is reduced there.

Observation. The network still follows the changing velocity input qualitatively, but the response is less clean than in Ex 2.4. In particular, the motion shows larger transient delays after velocity changes, and the measured response is weaker overall.

The hexagonal grid pattern is still visible in the centre of the network, but the bumps near the boundary are weaker because the input is attenuated there. This suggests that input attenuation still supports 2D path integration under aperiodic boundaries, although the tracking is less accurate than in the periodic case.

Exercise 4 — Final Question (Biology-Oriented)

Model predictions. The CAN model predicts that the hexagonal grid pattern emerges from recurrent centre-surround connectivity, and that its spatial phase is updated continuously by integrating velocity inputs. If the velocity modulation is removed (for example by setting $\alpha = 0$), the static grid can still exist, but it should no longer support path integration.

Proposed experiment. A simple way to test these predictions would be to record grid cells in medial entorhinal cortex (MEC) while manipulating the relationship between locomotion and self-motion cues. Rats could first explore normally so that grid cells are identified from their spatial firing pattern, and this would also provide a baseline measure of grid spacing and stability. Then, in a virtual-reality setup, the visual motion cues could be scaled relative to the animal's real movement by a factor κ .

If the CAN model is correct, the internal grid representation should drift according to the altered velocity signal rather than the true physical displacement. In other words, changing the gain should make the decoded neural position move too far or not far enough compared with the real position. By comparing neural position estimates with the true trajectory of the animal, one could test directly whether the MEC network updates position through velocity integration. A systematic error that grows over time under gain manipulation would be a strong sign that the network is integrating the modified self-motion signal.

A second test would be to perturb velocity-related inputs to MEC, for example by optogenetically weakening pathways carrying self-motion signals. The model predicts that this should disrupt grid updating more than the static grid structure itself. In that case, one might still observe a grid-like firing pattern, but its alignment with the animal's actual movement would become less accurate.

Conclusion. Overall, such experiments would test both key parts of the model: the recurrent origin of the grid pattern and the continuous updating of grid phase by velocity integration.

Conclusion

We have successfully implemented and analyzed a hierarchical path-integration model, progressing from a 1D ring attractor to a comprehensive 2D continuous attractor network (CAN) that recapitulates the spontaneous emergence and velocity-driven translation of hexagonal firing fields observed in MEC grid cells. Key findings include:

- The 1D Mexican-hat connectivity profile drives spontaneous pattern formation.
- A dual-population architecture structurally converts a scalar velocity signal into a proportional drift of the activity bump.
- Generalizing the architecture to 2D with four directionally shifted populations spontaneously generates a hexagonal lattice, representing the network's minimal-energy state.
- Applying a radial envelope $e(\mathbf{x})$ via input attenuation successfully restores the lattice in the network interior and strictly preserves the translation invariance required for undistorted dynamic path integration.

Approach

The team (Yann and Arthur) divided the simulation work into two parts. One implemented the 1D continuous attractor network for Exercises 0 and 1, while the other implemented the 2D extension and boundary-condition analysis for Exercises 2 and 3. After this first split, we systematically reviewed each other's code, reran all simulations together, and harmonized the notation, parameter choices, and plotting style across the notebook.

We wrote the simulation code in Python, using only the equations from the project statement and standard scientific libraries. For Exercises 0 and 1, we first coded the 1D Mexican-hat kernel, then the Euler integration of the ring attractor, then the two shifted subpopulations used for velocity integration, and finally the velocity readout. In Exercise 1.4, we chose to estimate the bump velocity through the phase of the first spatial Fourier component, because the bump lives on a periodic ring and its position is therefore naturally described as an angle rather than by a simple peak index. For each figure, we proceeded incrementally: first verifying that the kernel shape matched expectations, then checking that the firing-rate heatmaps reproduced the expected bump or periodic structure, and finally validating that the measured bump velocity scaled linearly with the imposed velocity.

For Exercises 2 and 3, we followed the same overall logic: we first generalized the kernel to 2D, then implemented the four directional populations and verified the emergence of a hexagonal grid, next tested the effect of constant and time-varying velocity inputs, and finally compared periodic and aperiodic boundaries together with the two correction strategies. For the moving-grid analysis, we kept the method as simple as possible by directly tracking the displacement of the activity pattern after an initial stabilization period. In the aperiodic-boundary part, we implemented both weight attenuation and input attenuation to compare a correction acting on the recurrent output of boundary neurons with one acting on their external drive. From these simulations, we retained that input attenuation is the more suitable choice for path integration, because it reduces edge effects while preserving the recurrent attractor structure more naturally.

We used a Large Language Model only as a support tool, not to generate the scientific content of the project. In practice, we used it to improve the English of the report, make parts of the code clearer and more readable, and fix minor syntax or plotting issues. Our prompts were therefore focused on code clarity, variable naming, and academic phrasing, rather than on asking for theoretical derivations or ready-made solutions. The modeling choices, debugging, interpretation of the figures, and final scientific conclusions were all done by the two of us.

References

- [1] T. Hafting, M. Fyhn, S. Molden, M.-B. Moser, and E. I. Moser, “Microstructure of a spatial map in the entorhinal cortex,” *Nature*, vol. 436, no. 7052, pp. 801–806, 2005.
- [2] Y. Burak and I. R. Fiete, “Accurate path integration in continuous attractor network models of grid cells,” *PLOS Computational Biology*, vol. 5, no. 2, p. e1000291, 2009.